Sushant Poudel

Computer Engineering · AI Systems & Security Research

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RESEARCH INTERESTS

Verification and safety of autonomous AI agents: whether a small, locally served model can act as the policy-enforcement layer that role-based access control cannot provide for an agent that executes actions rather than proposing them. Adjacent interests: tool-boundary and prompt-injection attacks on multi-agent systems, memory-safety and fuzzing of the libraries agents depend on, offline/edge inference under hard resource budgets, and networking for low-connectivity regions.

EDUCATION

Bachelor in Computer Engineering — Nepal Engineering College (NEC), Bhaktapur, Nepal
Final year, 8th semester | undergraduate researcher, Department of Computer Science and Engineering
Relevant coursework: machine learning, computer architecture, operating systems, computer networks, cryptography and network security, data structures and algorithms.

High School — Science (Physics, Mathematics) — St. Xavier's School, Jawalakhel
2020 – 2022 | GPA / percentage available on request

PUBLICATIONS & PAPERS

LLM Agent Firewall: Real-Time Detection and Neutralization of Prompt Injection in Multi-Agent Systems **Accepted · oral**
Paper ID 4005 — 2nd IEEE International Conference on Cybersecurity and AI-Based Systems (Cyber-AI 2026), Bucharest, Romania.

Evaluating Google's Open Knowledge Format (OKF v0.1) for Multi-Agent Threat Intelligence: A Proof-of-Concept in Digital Forensics **Accepted**
IC-AICSE 2026 — 7 October 2026, hybrid.

A Hybrid Causal-Generative Multi-Agent Framework for Secure Autonomous Mechatronics: Mitigating Real-Time Prompt Injections in Edge-Deployed Disaster Recovery Networks **Accepted**
Paper ID 1571313313 — with A. Pandey. 4th IEEE International Conference on Artificial Intelligence and Mechatronics Systems (AIMS 2026), presented virtually 23–24 July 2026; submitted to IEEE Xplore.

Edge-Native Semantic Firewall for Autonomous LLM Agents: A Structured Chain-of-Thought Verification Framework **Published**
S. Poudel, R. Pandey, A. Pandey — Department of Computer Science and Engineering, Nepal Engineering College.
PUBLISHED - Proceedings of the International Conference on Innovation in Computing, Science, Engineering and Technology (ICICSET), 2 October 2026. DOI: 10.65091/icicset.v3i1.75 | https://doi.org/10.65091/icicset.v3i1.75
A 3.8B-parameter model (Phi-3-mini, 4-bit quantized) running on a single consumer laptop inside a 4.2 GiB VRAM budget, evaluated on 600 policy scenarios per condition with no cloud inference. The central result is negative and contradicted our hypothesis: constraining output to JSON *without* a reasoning field produced the least safe evaluator of the three arms (46.2% unsafe accepts versus 17.2% unconstrained), while a mandated Chain-of-Thought field order cut false accepts of irreversible hard-denial commands from 71 to 6 and reached 66.3% decision accuracy. Residual failures are reported in the paper (6 of 208 hard denials still approved; roughly one verdict in ten changes between identical runs).
System, corpus, raw model outputs and camera-ready: github.com/sushant-me/Edge-Native_Semantic_Firewall_

PREBAS: Preemptive Bandwidth Scaling for WebRTC in LEO Satellite Networks Using Lightweight Neural Prediction **Accepted**
2026 IEEE RTC — Chicago, USA.
Real-time WebRTC video over low-Earth-orbit links breaks conventional congestion control: round-trip times are large and handovers drop the link without warning. The work uses a lightweight neural predictor to scale the send rate *ahead* of the degradation rather than reacting to it, evaluated against reactive baselines over emulated LEO conditions.

A Hybrid Causal-Generative Multi-Agent Framework for Secure Autonomous Mechatronics **Preprint, 2026**

An edge-native Causal Safety Enforcer that applies Pearl's do-calculus to intercept prompt-injection-driven actuation in disaster-recovery networks, with Byzantine-resilient swarm coordination and no cloud dependency.

Project Swayam — The Unbreakable Municipality **Conference paper, 2026**

Presented at the Municipal AI Governance Conference 2026 (Budhanilkantha Municipality, 14 June 2026). A fully offline sovereign e-governance stack: a quantized local model for assistance behind a deterministic gatekeeper that blocks data-extraction attempts.

EmbodiedOS: Offline-First Robotic OS with RAG Memory **Preprint, 2026**

Robotics stack combining C++ hardware control with local LLM memory for manipulation tasks, designed to operate with no network connection at all.

Direct-to-Device LEO Satellite Communication **Space Con 2026 — poster**

Architecture for 5G non-terrestrial-network connectivity from ordinary smartphones to LEO satellites, sized for Nepal's terrain where terrestrial coverage is uneconomic.

Designing Real-Time Verification Frameworks for Autonomous Agentic Decisions **Preprint, 2026**

A framework for bounding and actively verifying LLM-driven autonomous decisions in real time to prevent unsafe state transitions.

Evaluating Wi-Fi Sensing Attack Vectors: Covert Threat Detection and Mitigation **Preprint, 2026**

Offensive security research on covert surveillance through Wi-Fi Channel State Information, with deterministic mitigations for enterprise IoT deployments.

Nepal as a Data Center Hub of South Asia **Research paper, 2026**

Topological, climatic and energy-infrastructure analysis of Nepal as a sustainable data-centre location for the region.

RESEARCH & ENGINEERING EXPERIENCE

AI Engineer — Atmos SoftTech Pvt. Ltd.
Birgunj, Parsa, Nepal | March 2024 – March 2026 | full-time, 2 years

- Designed, developed and deployed AI-based systems and machine-learning models in production, owning features end to end from data preparation through deployment and monitoring.
- Built and maintained full-stack web applications — front-end interfaces, back-end services, databases, authentication and release automation — for company and client-facing products.
- Engineered autonomous agent systems: tool-calling pipelines, multi-step task orchestration, retrieval over private local data, and offline-first inference where sending data to a cloud API was not acceptable.
- Conduct, performance and professionalism rated highly satisfactory in the company's signed experience letter (Md. Samsad Ali, CEO), available on request.

Research Engineer — Arvion
Remote, part-time | from June 2026

- Continued employment confirmed after a three-month trial period, at a monthly stipend of NRS 22,500.

Technical Lead & Software Developer (Intern) — Avatar Tech Solutions Pvt. Ltd.
Kathmandu, Nepal | December 2025 – June 2026

- Architected, built and deployed the company's corporate web ecosystem and a responsive mobile application as the full-stack technical lead.
- Ran enterprise IT operations for 30+ workstations: setup, network configuration and zero-downtime maintenance.
- Built an automated inventory-management system mapping hardware serial numbers to generated reports, removing manual record-keeping from the process.

PEER-REVIEWED OPEN-SOURCE CONTRIBUTIONS

Work accepted or under review in Google's open-source libraries and agent frameworks. Every item below resolves to a public issue or pull request.

- **google/go-github — pull request authored and merged, then generalised by the maintainer.** My change stopped a release-asset upload being sent when the upload URL's host differed from the configured upload host; the maintainer replaced it the next day with a broader rule — *credentials are sent only to configured origins* — which is what master implements today and which carries the commit *"Address feedback from sushant-me"*. PR #4556 · PR #4564

- **google/s2geometry — five open memory-safety issues and two hardening pull requests.** A NULL dereference on the library's documented traversal path, an out-of-bounds read, an unchecked shape-id, and two denial-of-service allocations (16 GiB and 2.4 GiB) reachable from tiny malformed inputs, each with a standalone reproducer. PR #681 and PR #682 are under review by a Google maintainer. Two earlier fixes were closed, one of them because AddressSanitizer showed it fixed the wrong bug — the trace is in the thread.
- **google/libphonenumber.** The OSS-Fuzz as-you-type-formatter harness could never produce valid input and therefore never executed the code it targeted. The published OSS-Fuzz report shows 0.00%; a local rebuild of the same target measured 23.50% before and 93.05% after the fix. PR #4079
- **google/adk-go, adk-java, adk-python.** Fixes and reports for MCP tool shadowing: framework-reserved tool names that a server-supplied tool can occupy, bypassing the duplicate-name guard. #1606, #1515, #7145
- **Trail of Bits `skills`** and **sisakulint** — new detection rules for agentic GitHub Actions pipelines: unbounded tool targets and AI-action tool-pattern abuse.

Research software released

- **agentbound** — static detector for framework-side AI-agent tool-boundary bugs (tool shadowing, reserved-name collisions, last-wins tool dictionaries, fail-open confirmation gates); ten rules across Python, TypeScript and YAML. github.com/sushant-me/agentbound
- **mcp-nameguard** — checks an MCP server's tool names against the names agent frameworks reserve for themselves. github.com/sushant-me/mcp-nameguard
- **trajectorycheck** — trajectory-level evaluation for AI agents: grades tool calls, arguments, side effects and injected instructions, with a deliberately-broken agent as a control.
- **beyond-attention** — a selective state-space model (Mamba/S6) implemented from scratch in numpy, with a measurement harness that publishes the results which do not flatter the architecture.

SELECTED PROJECTS

GhostSignal — through-rubble life detection

Wi-Fi CSI · ESP32 · mesh networking · penetration-tested

Detects the chest movement of people trapped under rubble from Wi-Fi Channel State Information over a mesh of low-cost nodes — no camera, no wearable, works through debris.

Autonomous offline AI agent pipelines

Python · local LLMs · RAG · edge inference

Privacy-first agent stack that runs with no internet access, for continuous code generation and multi-agent task execution on local hardware.

VaultChat — steganographic secure channel

Flutter · end-to-end encryption · steganography

Encrypted peer-to-peer messaging concealed inside a functional scientific calculator, with a hidden partition unlocked only by cryptographic keycodes.

Arth-Sathi & Smart NEPSE — financial NLP

Python · LSTM · NLP · generative AI

Generative analyst for the Nepal Stock Exchange with an LSTM sentiment engine that re-ranks news sources by their historical predictive accuracy.

Dhukuti Pay — offline payment wallet

Flutter · local cryptography

Peer-to-peer wallet for secure transactions over local protocols with no active internet connection.

Mero-Bazaar-Secured — hardened marketplace

TypeScript · security review · CI

E-commerce platform taken through a pre-publication security review; 52 vulnerabilities closed before release.

TECHNICAL SKILLS

Research / ML PyTorch, TensorFlow/Keras, transformer and state-space architectures (Mamba/S6) implemented from scratch, LLM quantization and local inference (GGUF), RAG and LangChain, model evaluation and error analysis, causal inference fundamentals, NumPy/Pandas.

Security Prompt-injection and tool-boundary attacks on agent frameworks, MCP security, fuzz-harness engineering and coverage analysis, memory-safety triage (ASan/UBSan/GDB), static analysis (CodeQL, Semgrep), responsible disclosure.

Engineering Python, Go, Java, C/C++, TypeScript/JavaScript, Next.js and React, Node.js, Flutter/Dart, REST APIs, SQL, Docker and Kubernetes fundamentals, Linux, Git.

QA & automation PyTest, Jest, Cypress, Selenium, Postman, GitHub Actions CI/CD, TDD/BDD.

AWARDS & LEADERSHIP

- **Rank #1, HackingHub Q3 2026 global leaderboard** — 177 flags, 18,125 XP, one gold, two silver and two bronze awards; the next-ranked account holds 97 flags. Public source: api.hackinghub.io/leaderboard?type=previous

(Q3 board since rolled to Q4; the all-time board still carries these numbers).

- **Aspire Leaders Program — Global Finalist (2025).** One of ~10,000 finalists selected from 54,000+ applicants worldwide; leadership training co-created by Harvard Business School faculty.
- **Hult Prize — 1st Runner-Up, 2024 and 2025.** Consecutive recognition for social entrepreneurship.
- **Bagmati Province Hackathon 2026 — Runner-Up.**
- **NEC IT Club — Joint Secretary (2024–2025).** Logistics and administration for department-level technical events and hackathons; NEC Ingegium Hackathon volunteer organiser and competitor.
- **Conference participation:** Municipal AI Governance Conference 2026 (presenter); Pulchowk Hardware Competition (competitor).
- **NullOrigin CTF 2026** (international, 24-hour) — finished **2nd of the eight teams in the Grand Finale with team `roamers`: 16,850 points, 40 of 42 challenges solved** (final standings: 1st Jokers 17,600, 2nd roamers 16,850; window closed 25 September 2026); Grand Finale Certificate awarded 26 September 2026 (ID NO-2026-GF-229CF4, verifiable at creds.cyberhx.com)

PROFESSIONAL CERTIFICATES

IBM AI Engineering (IBM, Oct 2025) — deep learning with Keras/TensorFlow and PyTorch, generative AI and LLMs, transformer fine-tuning, RAG agents with LangChain. · **IBM DevOps and Software Engineering** (IBM, Mar 2026) — Agile, containers and Kubernetes, microservices, CI/CD, application security, observability. · **Google Cybersecurity** (Google, Apr 2025) — network security, Linux and SQL tooling, threat and vulnerability analysis, detection and response, security automation in Python. · **Google UX Design** (Google, Dec 2025) — user research, wireframing, prototyping.

Specializations: Deep Learning (DeepLearning.AI, Sep 2025); Machine Learning (Stanford University, Aug 2025); Flutter & Dart Complete App Development (Packt, Aug 2025). Plus 20+ further certificates from Meta, University of Michigan, Arizona State University and Scrimba — full list available on request.

References: academic and professional references available on request, including a signed experience letter from Atmos SoftTech Pvt. Ltd. · **Note:** publication statuses, leaderboard positions and upstream contributions are stated exactly as their public sources report them.